

University of Cape Town
University Examinations – June 2006
Department of Mathematics and Applied Mathematics
MAM2000W, 2002S, 2004H
Module 2LA : Linear Algebra

Time : 2 hours

Full Marks: 67

Marks available: 68

You may use only approved calculators.

*Answer **Section A** on the attached answer sheet, in the blocks provided, and **Section B** in the answer book provided.*

You may use the back of your answer book for rough work for Section A, but only the answers, not the rough work, will be marked. Number your answers to questions in Section B clearly, and present your answers clearly and coherently.

Make sure that you have filled in your student number on the answer sheet, detach it and put it inside your answer book before you hand it in.

This paper consists of 3 pages and an answer sheet.

Section A

1. Let

$$A = \begin{pmatrix} 2 & 0 & 0 \\ -2 & 4 & 2 \\ 2 & -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & -2 \\ -2 & 1 & 2 \end{pmatrix}.$$

- (a) Find $\det(A)$ and $NS(A)$.
- (b) Can A be written as a product of elementary matrices? Give a reason for your answer that does not depend on further calculations.
- (c) Find the rank and nullity of B (**not** A).
- (d) Give a basis for $NS(B)$ and a basis for $CS(B)$.
- (e) Find an upper triangular matrix C and elementary matrices E_1 and E_2 such that $B = E_1 E_2 C$.
- (f) Find all the eigenvalues of A .
- (g) For each eigenvalue, find a basis for the corresponding eigenspace.
- (h) Can A be diagonalized? If it can, give an invertible matrix that will diagonalize it; if not, say why not.

[2,1,2,3,4,3,4,2]

2. The following are subsets of P_3 :

$$A_1 = \{x^3 - 1, x^2 + 1, x^3 + x^2, x^3 + 2x^2 + 1\},$$

$$A_2 = \{x^3 - 1, x^2 + x, 1 - x, x^2\}, \quad A_3 = \{x^3 - 1, x^2 + x, 1, x^3\}.$$

In each case below, choose from the list above a set A_i ($1 \leq i \leq 3$) with the property given. You may use a set more than once, or not at all.

(a) A_i is a basis for P_3 .

(b) $\dim(\text{span}(A_i)) = 2$.

(c) $A_i \cup \{x^4\}$ is a basis for P_4 . [6]

3. Let $T : \mathbb{R}^3 \rightarrow M_{2 \times 2}$ be the linear transformation defined by

$$T \left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \right) = \begin{pmatrix} x_1 + 2x_3 & x_2 \\ -x_2 & x_1 + 2x_3 \end{pmatrix},$$

and let B and C be the bases of $\mathbb{R}^3$ and $M_{2 \times 2}$ given by

$$B = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}, C = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

(a) Find a basis for the kernel of T .

(b) Find the range of T and a basis for it.

(c) Which the following statements are correct?

(i) T is injective but not surjective.

(ii) T is surjective but not injective.

(iii) T is bijective.

(iv) T neither injective nor surjective.

(d) Find the matrix ${}_C[T]_B$ representing T with respect to the bases B (for $\mathbb{R}^3$) and C (for $M_{2 \times 2}$). [2,3,1,4]

4. Let V be an inner product space and $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$. You are given that

$$\langle \mathbf{v}, \mathbf{u} \rangle = -2, \langle \mathbf{v}, \mathbf{w} \rangle = 3, \langle \mathbf{u}, \mathbf{w} \rangle = -1, \|\mathbf{u}\| = 4, \|\mathbf{v}\| = 3, \|\mathbf{w}\| = 5.$$

Find $\langle \mathbf{v} + \mathbf{u}, \mathbf{u} + \mathbf{w} \rangle$ and $\|\mathbf{v} - 3\mathbf{w}\|$. [4]

Section B

5. Prove or disprove:

- (a) If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set, then $\{\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_3, \mathbf{v}_1 - 2\mathbf{v}_2 + 3\mathbf{v}_3\}$ is also linearly independent.
- (b) $S = \{A \in M_{2 \times 2} : \det(A) = 0\}$ is a subspace of $M_{2 \times 2}$.
- (c) The function $F : P_2 \rightarrow M_{2 \times 2}$ defined by $F(ax^2 + bx + c) = \begin{pmatrix} a+c & b \\ -b & 1 \end{pmatrix}$ is linear. [9]

6. Read the following result and its proof and then answer the questions following it.

Proposition A one-to-one linear transformation preserves linear independence.

Proof: Let V and W be vector spaces and $T : V \rightarrow W$ be a one-to-one linear transformation. Suppose $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a linearly independent set in V .

If $\lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) \cdots + \lambda_n T(\mathbf{v}_n) = \mathbf{0},$ (1)

then $T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \cdots + \lambda_n \mathbf{v}_n) = \mathbf{0},$ (2)

Hence $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \cdots + \lambda_n \mathbf{v}_n = \mathbf{0},$ (3)

From this it follows that

$$\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0 \quad (4)$$

and so the set $C = \{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$ is linearly independent. ■

- (a) Explain how (3) follows from (2).
- (b) What is used to deduce (4) from (3)?
- (c) Is the fact that T is a linear transformation used anywhere in the proof? If so, where and how is it used?
- (d) Let T and S be as in the proof. For each of the following statements, say whether it is true or false. If it is true, explain why it is true. If it is false, say what additional property T , or S , or both must have to make the statement true.
 - (i) C is a basis for $\text{ran}(T)$.
 - (ii) C is a basis for W .
 - (iii) $\dim(V) = \dim(\text{ran}(T))$.
- (e) Prove or disprove: Every linear transformation preserves linear independence. [11]

7. Prove or disprove:

- (a) There is a 2×2 matrix A such that $NS(A) = CS(A)$.
- (b) There is a 3×3 matrix A such that $NS(A) = CS(A)$.
- (c) If A is a square matrix with $NS(A) = CS(A)$, then A can be diagonalised. [7]